BABATUNDE ISAAC ROTIMI

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EDUCATION

PhD (In progress, 3rd Year), Geophysics and Seismology, Syracuse University, Syracuse, New York May 2023- Present
Advised by Dr. Joshua Russell, CGPA 3.78/4.0.

Graduate Coursework Completed: Sedimentary Basin Analysis, Geophysics, Numerical Methods Geosciences, Machine Learning in Earth Sciences, Geochemistry, Physical Hydrology, Plate Tectonics, Scripting for Data Analysis.

B.Sc. Applied Geophysics. CGPA 4.38/5.0 2017-2021
Adekunle Ajasin University Akungba Akoko (AAUA), Ondo State, Nigeria
Thesis: Seismic Interpretation of 2D lines from the OPL250 Deepwater, Niger Delta.

PROFESSIONAL/ RESEARCH EXPERIENCE

Research Assistant August 2023 – Present
Earth Imaging and Dynamics Laboratory, Syracuse University
Advisor: Dr. Joshua Russell.

- Developing Novel Hybrid Hierarchical Geophysical Inversion Engine for Basin scale Imaging.
- Subsurface Shear Velocity Characterization of Atlantic Coastal Plain Sediment Structure via incorporating HVSR, Rayleigh Wave Ellipticity, and Multimode Phase Velocities.
- Shallow Subsurface Imaging and Reservoir Characterization Using DAS/Nodal Array.

Research Affiliate Sept 2025 – Present
Mineral-X, Stanford University (Remote)
Supervisor: Dr. Peng Li.

- Uncertainty Quantification in Subsurface Imaging Via Physics Informed Neural Network coupled with Variational Inference.

Graduate Student Researcher January 2024 – June 2024
Basin Analysis Laboratory, Syracuse University
Advisor: Dr. Christopher Scholz

- Carried out 3D Seismic data interpretation of the Northeastern Rockall Trough Legacy data using OpendTect
- Mapped Faults and Performed Horizon Delineation of the most continuous horizons to produce distinct surfaces throughout the entire area.
- Produced Subcrop map of seismic facies of the depth surfaces and Performed Characterization of the underlying seismic facie using Amplitude, Frequency and Continuity
- Worked with a team of four to perform comprehensive structural features interpretation, volume construction and re-evaluation of the entire grid for hydrocarbon reservoir prospectivity and carbon sequestration.

Subsurface G&G Intern June 2019- August 2019
Danvic Petroleum International Lagos, Nigeria

- Conducted hydrocarbon evaluation of a Niger Delta field using seismic interpretation and attribute analysis.
- Estimated reservoir properties, including water and gas saturation, and evaluated oil-in-place volumes.
- Identified exploration leads and prospects through integrated subsurface analysis.

SELECTED PUBLICATIONS

Rotimi, B.I., J.B. Russell, A. García-Jerez. Shear-Wave Velocity Characterization of the Atlantic Coastal Plain Sediments via HVSR Inversion and Ambient Noise Dispersion. (in prep for JGR).

Rotimi, B.I., J.B. Russell, A. García-Jerez., Pratt, T., Fischer, K., Lin, F., Eilon, Z. Hierarchical Bayesian

Engine for Joint Inversion of HVSR and Surface Wave Data for Multi-scale Crustal Imaging of the Lower Appalachians & Atlantic Coastal Plain Sediments. (in prep for JGR).

SELECTED CONFERENCE PROCEEDINGS (* = talk, § = poster)

* **Rotimi, B.I.,** J.B. Russell, A. García-Jerez., Pratt, T., Fischer, K., Lin, F., Eilon, Z. Hierarchical Bayesian Engine for Joint Inversion of HVSR and Surface Wave Data for Multi-scale Crustal Imaging of the Lower Appalachians & Atlantic Coastal Plain Sediments. Seismological Society of America Meeting 2026, Pasadena, CA.

* **Rotimi, B.I.,** J.B. Russell, A. García-Jerez (2025). Shear-Wave Velocity Characterization of the Atlantic Coastal Plain Sediments via HVSR Inversion and Ambient Noise Dispersion. AGU Fall Meeting 2025, New Orleans, LA.

§ **Rotimi, B.I.,** J.B. Russell, A. García-Jerez (2024). Toward combining microtremor HVSR, Rayleigh wave ellipticity, and multimode phase velocities to constrain Atlantic Coastal Plain sediment structure. AGU Fall Meeting 2024, Washington, D.C.

FIELD WORKS

Earthscope Primary Instrumentation Centre, New Mexico Tech. 2024

- Broadband Seismometers Instrument Deployment Training, Socorro, New Mexico.

LIGO Hanford Site, Richland, WA. 2025

- One-week Broadband Seismometers Deployment Exercise

AWARDS AND FUNDING

Under Review:

- National Science Foundation- Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support 1 million Compute Credits ($7,000) Compute Credits, 2026
- GSA Student Research Grant, 2026

Awarded:

- National Science Foundation- Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support ACCESS-EES250003; 1.5 million ($10,000) Compute Credits (2025-01-14 - 2026-10-31)
- Recipient of Syracuse University Graduate Student Organization Travel Grant (2024,2025)
- Dr Igbalajobi E-Passport Grant (2022)
- Best Graduating Student, Applied Geophysics (2021)
- 2nd Runner Up, NAPE-YP Exciting World of Geosciences Virtual Quiz (2021)
- Award of Excellence for Outstanding Performance in Industrial Training Exercise (2020)
- Recipient of Ondo State Scholarship Board Assistance Grant (2019)

MISCELLANEOUS

Journal Referee: Society of Exploration Geophysicist; Earth and Space Science; Geochemistry, Geophysics, Geosystems

PROFESSIONAL AFFILIATIONS

- Member, American Geophysical Union (AGU)
- Fellow, Geological Society of London
- Member, New York State Council of Professional Geologists (NYSPG)
- Member, Society of Exploration Geophysicist (SEG)
- Member, Nigerian Association of Petroleum Explorationists (NAPE)
- Member, American Association of Petroleum Geologist (AAPG)

SKILLS AND COMPETENCES

Technical:

- **Languages**: Python, Matlab, R.
- **Frameworks**: Pandas, GeoPandas, NumPy, Scikit-Learn, Pytorch, Matplotlib.
- **Platforms**: PyCharm, Jupyter Notebook, Visual Studio, Git, HPC.
- Geological/Geophysical Field Surveys

Soft:

- Creative, Innovative, Flexible and enthusiastic team player with great leadership skills.
- Research-driven with keen data analytical abilities and the ability to listen attentively, communicate persuasively, and follow through diligently with assignments.